

HOW DO MODELS UNDERSTAND HUMAN VISION ILLUSION ON HOLISTIC PROCESSING

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ABSTRACT

The performance of machine vision has gradually improved in recent years and has even surpassed human vision. However, when attempting to understand the issue of human visual illusions, we need the model to have a higher alignment with humans. In the experiment, I first construct a Siamese architecture and try several different embedded models to test the recognition performance of the traditional visual model on two types of holistic-processing illusion pictures. After that, I utilize the CORnet-S model as embedded network, selects different time settings and explore their performance in the same task. Finally, I attempt to introduce the structure of HED into the CORnet-S model and tried to improve the CORnet-S model of 0-0-2 time setting. The experiments draw to the conclusion that high level vision significantly contributes to the holistic processing illusion, and more bionic and efficient model contents should be developed grounded on neurological evidence in future researches.

1 INTRODUCTION

Recent years, deep neural networks (DNNs) has reached gorgeous achievement in visual tasks. Computer vision has become increasingly important and effective in recent years due to its wide-ranging applications in areas as diverse as smart surveillance and monitoring, health and medicine, sports and recreation, robotics, drones, and self-driving cars. (Khan et al., 2018) Many scientist start to focus the difference and alignment between computer vision and human vision. However, the differences between these two visual systems decides that machine vision tends to be a mechanism using accurate computation methods to process input images, rather than a system fundamentally shaped by context, experience, and perceptual biases. This raises a critical question: do artificial vision models truly understand visual content the way humans do, or are they merely optimizing for performance on benchmark datasets?

One striking example is our susceptibility to visual illusions, where perception often diverges from physical reality. Visual illusions constitute strong examples of the dissociation between what is in the real world and what we end up perceiving. (Kreiman, 2021) Among the many types of visual illusions, holistic processing illusions, as one of the most common illusion, offers a unique perspective into how humans integrate facial information as a whole, rather than through isolated parts. We can see some ingenious design of models that realize this properties through structures or attention weights. However, how does this structures and designs help model focus on global information? And how does the final performance on holistic processing illusion like human vision, or align with human vision?

This project investigates these alignments and explore these performance by evaluating different embedding networks under a Siamese architecture on their responses to two well-known holistic illusions. To move beyond surface-level performance, we complement these behavioral results with structural and saliency analyses. We focus particularly on CORnet-S, a bionic model designed based on neurological evidence and test which temporal recurrence structure contributes to alignment. Furthermore, we explore a hybrid extension that integrates edge-based processing (HED structure) to examine whether low-level visual features can enhance model-human similarity. Through these ex-

periments, we aim to provide a more mechanistic understanding of how architectural and functional constructions influence a model's ability to simulate human vision illusion on holistic processing.

2 HYPOTHESIS

We hypothesize that current computer vision models are not susceptible to human visual illusions, showing low alignment in terms of behavioral performance, saliency attention, and architectural structure. The bionic model CORnet-S when configured with proper time settings can show better alignment compared to other models. Integrating side-processing structures inspired by HED can help to improve the alignment.

3 LITERATURE REVIEW

Human and Machine Vision Alignment Inspired by a lecture from Dr. Serre, I became interested in the problem of alignment between computer vision and human perception. Dr. Serre and colleagues have shown that current deep neural networks often fail to replicate the decision processes underlying human behavioral judgments (Fel et al., 2022). In particular, the final layers of CNNs tend to focus on features that differ from those emphasized by humans (Serre, 2019). This project builds upon these insights, aiming to understand and improve the alignment between human and machine vision.

Methods to Improve Alignment Serre et al. modified DNN training using human attention data from ClickMe (Linsley et al., 2017), enabling the model to exhibit results consistent with human visual representation. (Linsley et al., 2023) This work suggests that structural modifications may also enhance the alignment between human and artificial vision systems, inspiring us to modify model structures.

Bionic Model – CORnet-S CORnet-S is a shallow recurrent neural network with four anatomically mapped areas and recurrent connectivity, which is the top model on Brain-Score and outperforms similarly compact models on ImageNet. (Kubilius et al., 2019) Its visual-cortex-resembled design makes it a suitable candidate for testing how recurrence depth affects alignment between human and machine vision. The details of CORnet model will be introduced in the following contents.

The Alignment Field – Illusion Visual illusions serve as compelling evidence of the disconnect between the physical world and our perceptual experience. (Kreiman, 2021) Unfortunately, many current vision-language systems fail to interpret these illusions accurately, often yielding misleading conclusions. (Ullman, 2024) This highlights illusion as a prominent domain where the alignment between human and machine vision remains notably weak.

Why Holistic Processing The domain of visual illusions is vast and highly diverse, with different types of illusions exhibiting distinct and often incompatible characteristics. Prior research in related areas has made significant strides: Dr. Serre and colleagues has introduced a recurrent network model of classical and extraclassical receptive fields that is constrained by the anatomy and physiology of the visual cortex explains a lot about the contextual illusion. (Mély et al., 2018) They also constructed γ -net, a model that explains orientation-tilt illusion. (Linsley et al., 2020). As holistic processing plays a crucial role in many visual illusions, improving its alignment with human perception is essential. This project aims to address that gap by enhancing model alignment in the context of holistic visual understanding.

4 METHODS

To investigate the alignment between human and machine vision in the context of holistic processing illusions, we employ a Siamese network architecture to determine whether a given pair of face images belongs to the same individual. Following training on a standard face distinguishing task, we evaluate the model's behavior on specially designed illusion images. If the model classifies the two illusion images as originating from the same individual, this is interpreted as a sign of high alignment with human perceptual processing. Conversely, if the model identifies the illusion images as belonging to different individuals, this suggests a lower degree of alignment with human vision.

4.1 MODEL STRUCTURE

Dataset The dataset used for training is `lfwcrop_color` <https://conradsanderson.id.au/lfwcrop/>, which is a cropped version of the Labeled Faces in the Wild (LFW) dataset (Huang et al. (2008)), retains only the central facial region and the background is omitted. For each LFW image, the area inside a fixed bounding box was extracted. The bounding box was at the same location for all images, with the upper-left and lower-right corners being (83,92) and (166,175), respectively. The extracted area was then scaled to a size of 64x64 pixels. The selection of the bounding box location was based on the positions of 40 randomly selected LFW faces. (Sanderson & Lovell, 2009) In this project, the dataset is labeled, offered by the curators of the website and dataset, to show whether two images come from same person or not.

Siamese Network To compare the pairs of images, we adopt a Siamese network architecture (Bromley et al., 1993). This architecture comprises two embedding networks with the same structure and shared weights, which are used to extract the features of the input signature and determine the similarity of the signature by calculating the distance between the feature vectors. The Siamese network paradigm has served as a foundational framework for various tasks such as face verification, image matching, and one-shot learning.

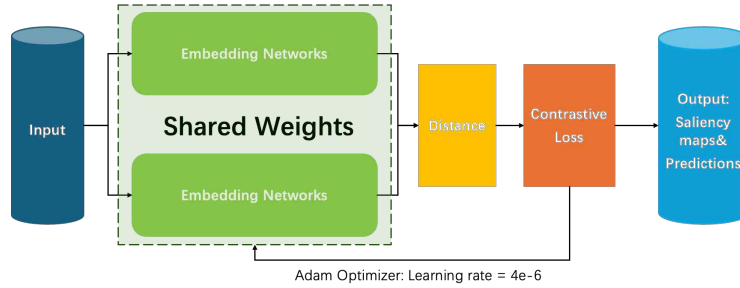


Figure 1: Siamese Network Structure

Adam Optimizer The Adam optimizer is utilized to train the Siamese network. (Kingma & Ba, 2017) Adam combines the advantages of both momentum and RMSProp by maintaining adaptive learning rates for each parameter through first and second moment estimations. Compared to vanilla SGD, Adam offers better stability and faster convergence in contrastive learning tasks, particularly under unbalanced gradients from positive and negative pairs. In Siamese networks where embeddings are learned from pairwise inputs with contrastive loss, Adam helps efficiently balance updates and stabilize training with minimal hyperparameter tuning.

Contrastive Loss The contrastive loss is used to compare the distance of two images and acts as objective function. The contrastive loss (Hadsell et al., 2006) is defined as:

$$\mathcal{L}(y, D) = y \cdot D^2 + (1 - y) \cdot \max(0, m - D)^2, \quad (1)$$

where $D = \|f(x_1) - f(x_2)\|_2$ is the Euclidean distance between the two feature vectors, $y \in \{0, 1\}$ is the binary label, and m is a predefined margin. Here the value of margin is 1.0 and the Euclidean distance is defined as the L2 norm distance between two vectors:

$$\|a - b\|_2 = \sqrt{\sum_i (a_i - b_i)^2} \quad (2)$$

4.2 EMBEDDING NETWORKS

ResNet50 Residual Network (ResNet) is an architecture that solves the difficult problem of training deep neural networks by introducing residual connections. (He et al., 2016) It is widely used in machine vision tasks, such as image classification, object detection, and feature extraction, and is the basic backbone network of multiple mainstream vision models. Here we implement pretrained ResNet50 networks <https://docs.pytorch.org/vision/0.18/models/generated/torchvision.models.resnet50.html> as the embedding networks of Siamese architecture to show its performance on Holistic Processing illusion test. To apply it in Siamese network, the final fully connected classification layer is removed..

ViT The Vision Transformer (ViT) is the first successful application of the Transformer architecture to image classification tasks. (Dosovitskiy et al., 2021) It partitions an image into fixed-size patches and treats them as tokens, enabling a different global modeling approach than CNNs. ViT achieves comparable or better performance on large-scale data and complicated visual task. Pretrain ViT network of timm (Wightman, 2019) is used as the embedding networks of Siamese network to show its performance on Holistic Processing illusion test. To apply it in Siamese network, the classifier head is deleted.

HED Holistically Nested Edge Detection (HED) is a deep learning edge detection method, which proposes to optimize both local and global edge features through multi-scale and deep supervision mechanism, and achieves better performance than traditional methods. (Xie & Tu, 2015) Its structure is usually based on VGG16 (Simonyan & Zisserman, 2015), and it uses multiple side-outputs for edge fusion, which is of great value in image understanding and human perception modeling. In this project, a HED structure is reconstructed with the pretrained VGG16 block <https://docs.pytorch.org/vision/main/models/generated/torchvision.models.vgg16>, which is applicable for Siamese Network to embed.

CORnet-S As shown in literature reviewed 3 and following figure 2, the CORnet-S is a bionic model which is structurally highly aligned to human brain visual cortex. To apply in the Siamese network, the classifier layer of the standard CORnet-S model is removed. In this project, the following time settings 2 of the recurrent time of each layer are explored to test the CORnet performance and each layer's function. These recurrent processes simulate the gradual accumulation and processing of visual information in the visual cortex.

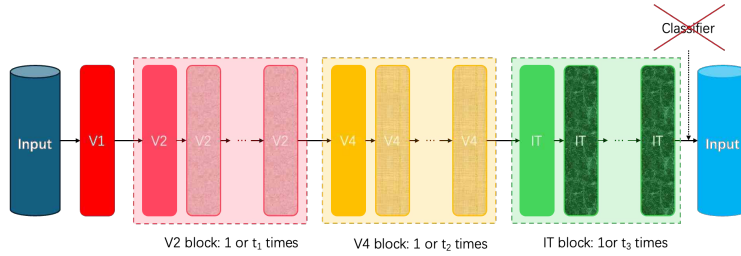


Figure 2: CORnet-S Network Structure

Table 1: CORnet-S recurrent time settings (V2, V4, IT)

Time Settings	V1 layer	V2 layer	V4 layer	IT layer
0.0_1	1	1	1	1
0.0_2	1	1	1	2
0.0_4	1	1	1	4
0.2_1	1	1	2	1
0.2_2	1	1	2	2
0.2_4	1	1	2	4
2.4_1	1	2	4	1
2.4_2	1	2	4	2
2.4_4	1	2	4	4
5.10_5	1	5	10	5

Hybrid CORnet-S To improve the alignment and performance of CORnet-S (Time setting: 0.0.2) with less recurrence, avoiding too much recurrent time and too deep networks, which reduce the structural alignment, HED structures are integrated into this CORnet-S model. Side Convolution preserves the shallow edge or structure information at different scales, which helps the model integrate multi-layer features. Similar to the lateral path of U-Net or FPN, the contribution of low-level details to the final expression is improved. Feature Fusion Convolution plays the role of unifying dimensions and integrating multi-scale features. Attention weights learns a channel attention weight

(which can be thought of as a gating mechanism) over multiple sources. Attention weights can emphasize the collaborative contribution of the temporal integration semantics and fused features of the "IT layer" to improve the discrimination ability.

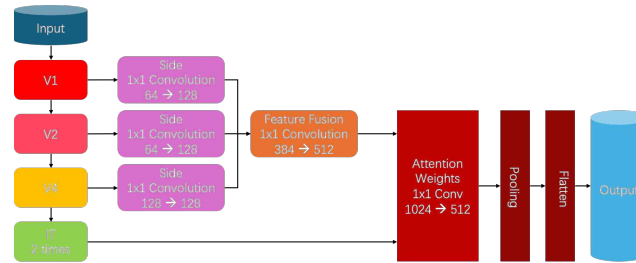


Figure 3: Hybrid CORnet-S Network Structure

4.3 TEST

Alignment Criteria Holistic illusion images consist of visually distinct inputs that, due to the nature of human holistic perception, appear almost indistinguishable. In this study, alignment in the domain of holistic processing is defined as a model's greater reliance on global facial structure rather than on local or isolated features. From a behavioral perspective, a model that focuses primarily on local details is likely to distinguish between illusion image pairs, demonstrating low alignment with human perception. Conversely, a model that fails to distinguish between such image pairs, as humans typically do, is considered to exhibit high perceptual alignment. From a structural standpoint, a model is regarded as aligned if its architecture contains resemblance to the hierarchical organization of the human visual cortex.

Thatcher Illusion Upside-down faces are hard to recognize because of the loss of facial expression in such faces. (Köhler, 1960) The eyes and mouth convey most information about an individual's mood and it is these areas "which can assist us in comprehending what they are trying to communicate to us". (ELLIS, 1975) According to these researches, a illusion, using Ms. Margaret Thatcher and other people's faces, but inverting their eyes and then flipped the whole images vertically, has created and named after Thatcher Illusion. (Thompson, 1980) In this context, we use the Thatcher Illusion to assess each model's ability to detect subtle local feature changes under perceptually challenging conditions, where humans typically fail due to holistic processing. According to the criteria in Section 4.3, models that successfully differentiate between Thatcher Illusion images likely rely on local features and exhibit weak holistic processing. In contrast, models that fail to distinguish them, successfully imitating human limitations, are considered to demonstrate strong holistic alignment. Two sets of Thatcher Illusion image pairs are used in this evaluation.

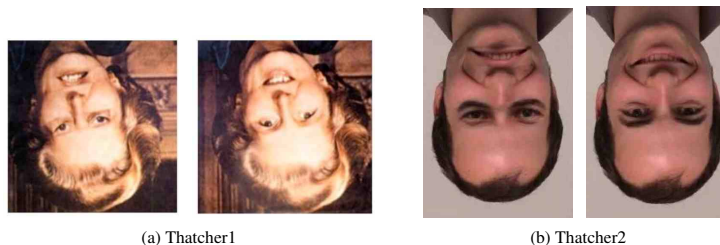


Figure 4: Thatcher Illusion

Part-Whole Illusion The Part-Whole Illusion is grounded in the finding that individual facial features—such as the eyes, nose, or mouth—are more easily recognized when presented within the context of a complete face. This phenomenon was first demonstrated by Sunday et al. (2017) in an experiment where participants were asked to memorize a full face and subsequently identify a

specific part, either in isolation or embedded within the whole face. Results indicated that participants performed significantly better when the target feature was presented in the context of the full face, supporting the notion that face recognition relies on holistic processing mechanisms. Based on this principle, a set of Part-Whole Illusion images is constructed in which subtle modifications were made to either the mouth (left image) or the eyes (right image). Although the images are objectively different, they appear strikingly similar due to the strong integration of facial features in human perception. Human observers can detect these differences, but doing so typically requires additional time and effort. Following the alignment criteria outlined in Section 4.3, a model that easily distinguishes between the modified image pairs likely relies on localized feature analysis and thus exhibits relatively weak holistic processing. In contrast, a model that fails to discriminate between them may be more reliant on global facial structure and therefore more closely aligned with human perceptual strategies. In this study, we evaluate two sets of Part-Whole Illusion image pairs to assess each model's capacity for holistic processing.

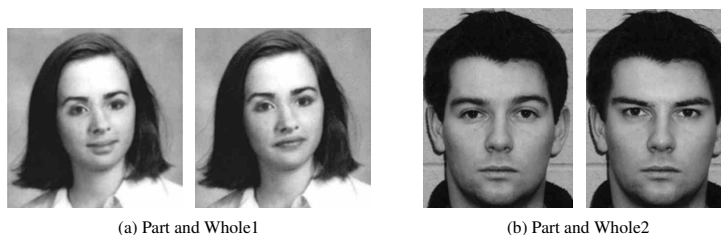


Figure 5: Part and Whole Illusion

Tests Claim The significant difference between Thatcher illusion and Part-Whole Illusion is that Thatcher illusion is hard for human vision to discriminate while the Part-Whole Illusion can be discriminated by human vision though some difficulties. This infers that the model are not expected to perform extremely deterministic result to show that two images are from the same person as what we expect on the Thatcher test.

Prediction Calculation All the prediction calculation are unified with following algorithm when predict the result of Siamese network with five embedding networks. I mapped the Euclidean Distance to a probability using a parameterized sigmoid function:

$$P_{\text{similarity}} = \sigma(-\alpha \cdot D + \beta) \quad (3)$$

Here D represents the Euclidean Distance and is calculated by L2 norm 2. The hyperparameter α is set to 10 and β is set to 5. The final outputs include the Euclidean distance, the similarity probability, and a binary decision, with visualizations provided to interpret the model's response of Thatcher Illusion and Part-Whole Illusion. We recognize the similarity over 50% to be the sign that the model cannot tell the difference between two illusion images.

Saliency maps To interpret model attention, we generate Saliency maps by computing input gradients with respect to the model's output norm. Specifically, we forward an input image $\mathbf{x}$, which is selected from the very first 9 or 10 validation set images¹, through the network and compute the output embedding $\mathbf{f}(\mathbf{x})$. The scalar score $\|\mathbf{f}(\mathbf{x})\|$ is then backpropagated to obtain the gradient $\nabla_{\mathbf{x}}\|\mathbf{f}(\mathbf{x})\|$, indicating how sensitive each input pixel is to the output. The saliency map is obtained by taking the element-wise maximum over the absolute gradient across channels. We visualize only the top 10% most salient pixels by showing them as red points. If the pixel is more salient, then the pixel is more heavily red. This provides an intuitive understanding of where the networks focus more, which offers side reflective evidence indicating models' holistic processing ability.

Code Explanation In the Github repository <https://github.com/LichanghengXJTU/NE240BAI-HolisticProcessing-Illusion.git>, there is a directory called Holistic Processing Illusion Codes, within there are five .ipynb documents and five .py documents. Here are their names and functions.

¹The index 4 and index 5 images in validation set are the same, so when I show the saliency map, sometimes I only show the first 9 pictures, but sometimes I show the first 10 pictures

Table 2: Github Documents

Name	Funtion
HP_ ResNet.ipynb	The ResNet50 embedded model and output in Jupyter Note book
hp_ resnet.py	The ResNet50 embedded model in python
HP_ ViT.ipynb	The ViT embedded model and output in Jupyter Note book
hp_ vit.py	The ViT embedded model in python
HP_ HED.ipynb	The HED embedded model and output in Jupyter Note book
hp_ hed.py	The ResNet50 embedded model in python
HP_ Cornet_ S.ipynb	The CORnet-S embedded model and output in Jupyter Note book
hp_ cornet_ s.py	The CORnet-S embedded model in python
HP_ HybridCornet.ipynb	The Hybrid CORnet-S embedded model and output in Jupyter Note book
hp_ hybridcornet.py	The Hybrid CORnet-S embedded model in python

Mention: The HP_ ViT .ipynb cannot be previewed in the github due to some mysterious reasons. But it can be normally downloaded. The other documents are all normal. The python documents are shown in case of corresponding notebook document cannot be viewed, which have the same code with the notebook document.

The output images are in the `output` directory and allocated in different directories according to each embedding networks' name. Readers can find training curves, epoch data, saliency maps, model parameters summary, prediction histogram chart and other data figures or pictures.

5 RESULTS

Summary of The Result As summarized in Table 3, we observe that only the CORnet-S model with time settings 0_0_4 and 0_0_2 consistently exhibit both behavioral and structural alignment with human holistic processing.

Table 3: Performance and Alignment Comparison Across Models

Model	T1 (%)	T2 (%)	PW1 (%)	PW2 (%)	Sal Align?	Struct Align?
ResNet50	98.75	98.89	99.01	98.28	✗	✗
ViT	0.00	81.90	78.80	88.74	✓	✗
HED	90.00	97.79	97.50	95.41	✗	✗
CORnet-S (0_0_4)	91.20	71.10	97.90	98.50	✓	✓
CORnet-S (0_0_2)	62.40	58.80	98.00	89.00	✓	✓
CORnet-S others	—	—	—	—	✓	✓
Hybrid CORnet	0.10	26.70	51.40	63.10	✓	✓/✗

The followings are specific result and analysis of each embedded models.

ResNet50 With figure 6, the ResNet50 embedded network cannot tell the difference between Thatcher illusion images and Pair and Whole Illusion images. However, this performance cannot be interpreted as ResNet understanding human vision on the aspect of holistic processing. This is because from the saliency map, the ResNet 50 focus on the most significant features of human faces, whom does not look human faces from a globally considering side. This shows that ResNet50 performs as understanding human vision because it is a too deep network, but its structure does not show its alignment of human vision.

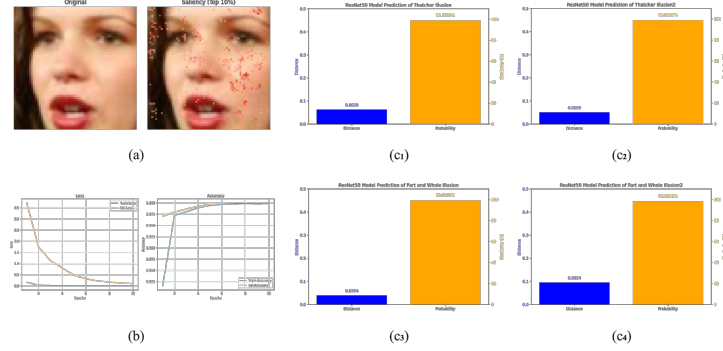


Figure 6: Result of ResNet50 Embedded Network. (a) The fourth-generated saliency map, showing the model's attention focused on the cheek and other large regions of the face. (b) The training curve of the model. (c) The model's predictions on various illusion images: (c1) Recognizes the Thatcher1 illusion images as 98.7520% the same person. (c2) Recognizes the Thatcher2 illusion images as 98.8919% the same person. (c3) Recognizes the Part and Whole1 illusion images as 99.0103% the same person. (c4) Recognizes the Part and Whole2 illusion images as 98.2811% the same person.

ViT With figure 7, the ViT embedded network can tell the difference between Thatcher illusion images group 1, but cannot tell the difference between Thatcher illusion images group 2 and Pair and Whole Illusion images. I interpret this phenomenon as: For Thatcher illusion, the ViT holistically understand it. Although it shows 81% probability that the two people are the same, this is much lower than the probability in the same case offered by other models (ResNet50: 98%, HED: 98%). And the first group of Thatcher illusion is the earliest and most used illusion images, which is more representative than the second group of images. For Part and Wole illusion images, the high similarity because ViT can recognize the difference between the illusion images, while human can also achieve this but with difficulties 4.3. So, these tests shows that ViT has relatively weak ability in Holistic Processing. And considering the structure of ViT, which is based on Transformer architecture and has an enormous of parameters and a large size, ViT has low alignment with human vision which has layer structures.

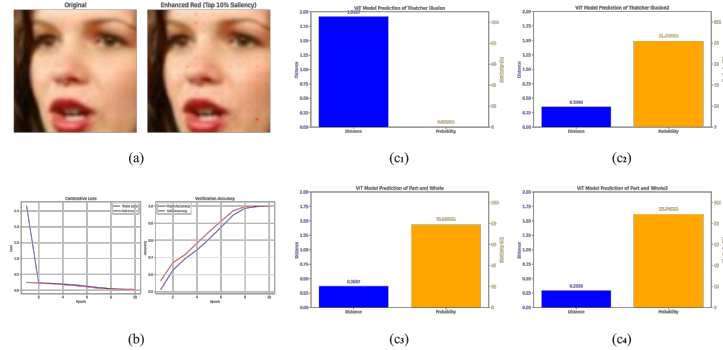


Figure 7: Result of ViT Embedded Network. (a) The fourth-generated saliency map, showing the model's attention is really distracted, considering the global information with focus. (b) The training curve of the model. (c) The model's predictions on various illusion images: (c1) Recognizes the Thatcher1 illusion images as 0.0001% the same person. (c2) Recognizes the Thatcher2 illusion images as 81.9002% the same person. (c3) Recognizes the Part and Whole1 illusion images as 78.8050% the same person. (c4) Recognizes the Part and Whole2 illusion images as 88.7400% the same person.

HED With figure 8, the ResNet50 embedded network cannot tell the difference between Thatcher illusion images and Pair and Whole Illusion images. From the saliency map, we can see that the HED model's attention is much more distracted compared to ResNet50 model, but focus on the most representative edges on human faces: eyes, noses, mouths, wrinkles, etc. This can be interpreted has the HED model combines the edge features and latent features of human faces, which is intuitively aligned with human vision. However, comparing the two visions' structure, HED focuses more on the low level vision (edges), which is not aligned with human vision.

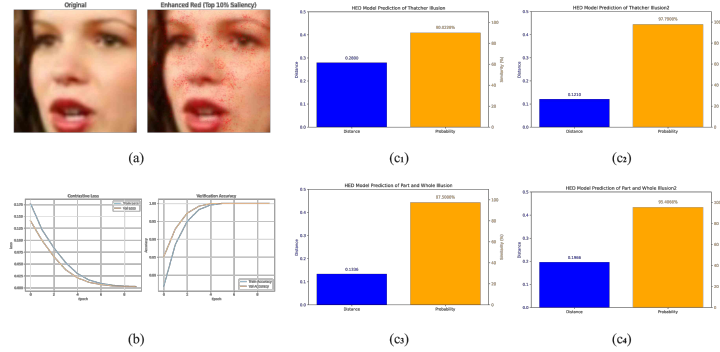


Figure 8: Result of HED Embedded Network. (a) The fourth-generated saliency map, showing the model's attention is really distracted, considering the global information with focus. (b) The training curve of the model. (c) The model's predictions on various illusion images: (c1) Recognizes the Thatcher1 illusion images as 90.008% the same person. (c2) Recognizes the Thatcher2 illusion images as 97.790% the same person. (c3) Recognizes the Part and Whole1 illusion images as 97.506% the same person. (c4) Recognizes the Part and Whole2 illusion images as 95.406% the same person.

CORnet-S With Figure 9, the CORnet-S model with time settings 2_4_2, 2_4_4 and 5_10_5 cannot be converged under current training settings. The CORnet-S model with time settings 2_4_1, 2_4_2, 2_4_4 and 5_10_5 shows low validation accuracy. In this case, these are called "inapplicable settings", while the other six models are "applicable settings". And only the "applicable settings" are considered. From the four prediction figures, the CORnet-S model with 0_0_4 performs the best: recognizes the Thatcher1 illusion images as 91.2% the same person, recognizes the Thatcher2 illusion images as 71.1% the same person, recognizes the Part and Whole1 illusion images as 97.9% the same person, recognizes the Part and Whole2 illusion images as 98.5% the same person. Among other models, only CORnet-S model with 0_0_2 performs alignment with human vision: it recognizes the Thatcher1 illusion images as 62.4% the same person, recognizes the Thatcher2 illusion images as 34.8% the same person, recognizes the Part and Whole1 illusion images as 98.0% the same person, recognizes the Part and Whole2 illusion images as 89.0% the same person. This shows that the CORnet-S model with 0_0_4 and 0_0_2 cannot tell the differences between the illusion images and can understand human vision illusion in holistic processing, and CORnet-S model with 0_0_4 performs more alignment with human vision than CORnet-S model with 0_0_2.

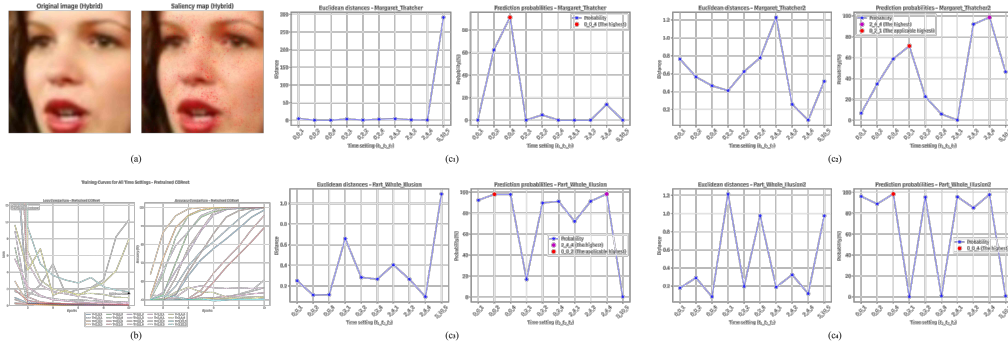


Figure 9: Result of CORnet-S Embedded Network. (a) The fourth-generated saliency map of the CORnet-S model with time setting 0_0_4, showing the model's attention is distracted, considering the global information. (b) The training curve of the models with different time settings. (c) The model's predictions on various illusion images: (c1) The prediction of CORnet-S model with 10 time settings on Thatcher1 illusion images. (c2) The prediction of CORnet-S model with 10 time settings on Thatcher2 illusion images. (c3) The prediction of CORnet-S model with 10 time settings on Part and Whole1 illusion images. (c4) The prediction of CORnet-S model with 10 time settings on Part and Whole2 illusion images.

Hybrid CORnet With figure 10, the Hybrid CORnet model, which designed to improve the CORnet-S model with 0_0_2 time settings' performance, performs worse than CORnet-S models with 0_0_2 and 0_0_4 time settings. It recognizes the Thatcher1 illusion images as 0.1% the same person, recognizes the Thatcher2 illusion images as 26.7% the same person, recognizes the Part and Whole1 illusion images as 51.4% the same person, recognizes the Part and Whole2 illusion images as 63.1% the same person. This shows that the Hybrid CORnet model can tell the differences be-

tween Thatcher illusion images, but cannot tell the difference between Part and Whole images. And the similarities offered by Hybrid CORnet on Part and Whole illusion images are lower than the two CORnet-S models. This indicates that the Hybrid CORnet model have low alignment with human vision on holistic processing, especially compared with CORnet-S models with time settings 0_0_4 and 0_0_2. The method to join HED side structure to CORnet-S model is failed.

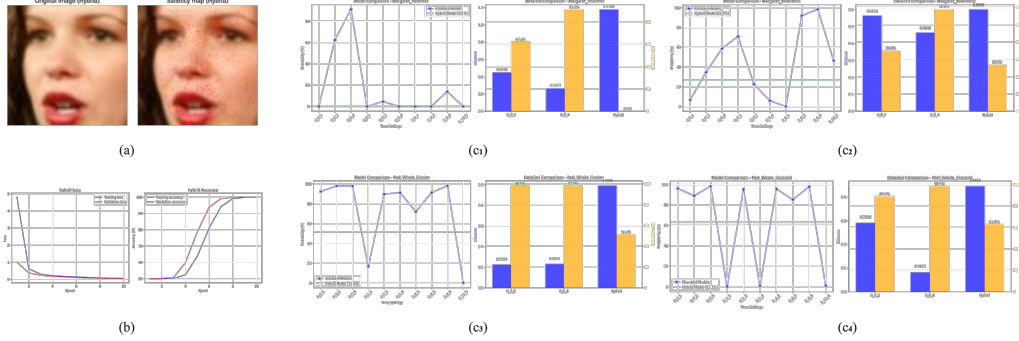


Figure 10: Result of Hybrid CORnet Embedded Network. (a) The fourth-generated saliency map of the Hybrid CORnet model which modified the CORnet-S model with time setting 0_0_2, showing the model's attention is distracted, considering the global information. (b) The training curve of the Hybrid CORnet model. (c) The Hybrid CORnet model and other CORnet-S models prediction comparison on various illusion images: (c1) The prediction comparison of Hybrid CORnet model and other CORnet-S models on Thatcher1 illusion images. (c2) The prediction comparison of Hybrid CORnet model and other CORnet-S models on Thatcher2 illusion images. (c3) The prediction comparison of Hybrid CORnet model and other CORnet-S models on Part and Whole1 illusion images. (c4) The prediction comparison of Hybrid CORnet model and other CORnet-S models on Part and Whole2 illusion images.

6 DISCUSSION

Interpret for non-bionic model Due to the criteria of alignment 4.3, the models are expected to show alignment both in performance and structures. ViT 5 shows low alignment in this task, while ResNet50 5 and HED 5 shows high alignment with human vision. However, all of three embedded networks are not structurally aligned with human vision cortex. Finding bionic model and observe its performance is the key point finding alignment between human vision and machine vision.

IT recurrence interpretation From the result of CORnet-S models with 10 different time settings 5, we find out two properties:

- 1. Models with fewer recurrences in low-level blocks (V_1 , V_2 , V_4) perform well in both original and illusion tasks.
- 2. Models with more recurrences in the high-level block (IT) perform well in illusion tasks.

This can draw to the conclusion that the low level vision is necessary but only too much recurrence in low level vision will bring bias to the input images and cause the not ideal performance. And with finitely-more IT block, which belongs to high level vision, the model performs better on the illusion task. This shows that IT is very probably to be the key cortex region that contributes to human vision illusion.

Hybrid is a failed test Looking HED performs high alignment with human vision, I firstly thought whether HED structure can help to improve the model's performance, substituting some of the IT recurrent time. This thought is based on the logic that more joint weights focusing on the low level vision can make up for the high level vision recurrence, which is also HED's features 5. The result of Hybrid proves that this logic is wrong. 5 Focusing on the low level vision does not help the illusion recognition. To reduce the recurrent time of IT block, we may need to promote the IT block structure which bionically imitate the IT cortex with neurological evidence rather than simply using the ResNet18 backbone.

Number of parameters Considering the brain structures of vision cortex, I am looking for not complicated and less parameter models, especially in the bionic models (CORnet-S and Hybrid CORnet). Only the embedded networks' trainable parameters are considered.

Table 4: Model Structure Metrics

Model Name	Number of Parameters (Trainable)	Number of layers
ResNet50	24,032,576	50
ViT	85,798,656	\
HED	14,780,352	19
CORnet-S 0_0_1	11,689,512	17
CORnet-S 0_0_2	52,069,928	18
CORnet-S 0_0_4	52,088,360	20
CORnet-S 0_2_1	14,215,208	18
CORnet-S 0_2_2	54,595,624	19
CORnet-S 0_2_4	54,614,056	21
CORnet-S 2_4_1	14,851,880	21
CORnet-S 2_4_2	55,232,296	22
CORnet-S 2_4_4	55,250,728	24
CORnet-S 5_10_5	55,277,224	34
Hybrid CORnet	52,827,816	22

Based on the goal of constructing Hybrid CORnet: to improve the alignment and performance of CORnet-S with less recurrence, avoiding too much recurrent time and too deep networks, we find out that the Hybrid model is even deeper and with more parameters than the CORnet-S model with time setting 0_0_4. The Hybrid model does not improve the performance with simplified structure, but using even complicated model to perform worse. So this also confirms that the proper way to improve the alignment between human vision and machine vision is to promote the IT block's structure, rather than making the model focusing on low-level vision.

Two kind of illusion comparison As Hybrid model result shows, 5, Hybrid model cannot tell the difference between Part and Whole illusion images, but can tell the difference between Thatcher illusion images. This may tell us the neurological functions of human brain taking holistic processing task are different in somewhat degree on processing Thatcher illusion and Part and Whole illusion. For the same, this may inspire us interpretation of this phenomenon should be taken with further discovered neurological evidence.

7 IMPLICATION

Finding Neurological Evidence This study concludes that the inferotemporal (IT) cortex plays a significant role in mediating holistic processing illusions, and that different types of holistic illusions may engage distinct neural mechanisms. These insights are derived from model-based analyses and empirical performance comparisons. To validate and further support these findings, future research should seek corroborating evidence from neuroscience—particularly through neuroimaging or electrophysiological studies that examine the functional roles of IT and other visual cortical areas in processing holistic visual information.

Future Implementation While machine vision models continue to advance rapidly, their utility is ultimately constrained by how well they align with human perceptual experience. Visual illusions exemplify a fundamental divergence between machine and human vision—not because such illusions exist inherently in the world, but because they emerge from the cognitive and perceptual mechanisms of human observers. In certain contexts, such as driving, these illusions can lead to serious consequences, including accidents or misjudgments. To enhance the societal value and safety of machine vision systems, it is essential that models incorporate an understanding of the perceptual biases and limitations inherent in human vision. Future development of vision models should thus be informed not only by engineering performance metrics, but also by interdisciplinary collaboration between machine learning researchers and neuroscientists. A more human-aligned and cognitively informed modeling approach will contribute to the creation of systems that are both functionally robust and perceptually aware.

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A PREDICTIONS

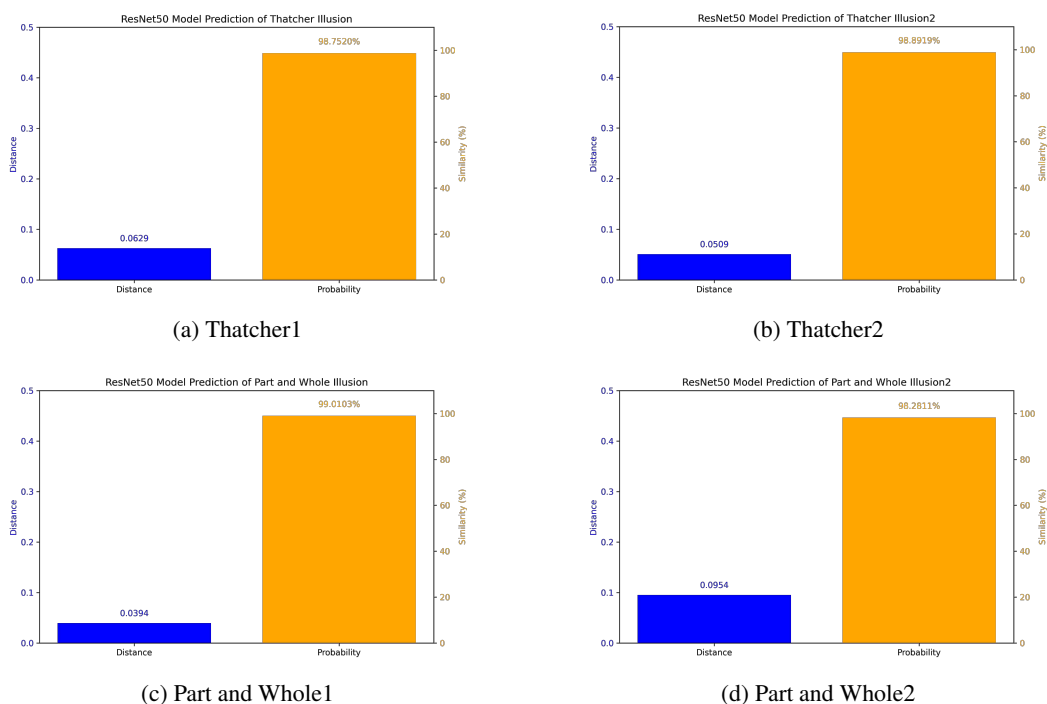


Figure 11: Prediction Result of ResNet50

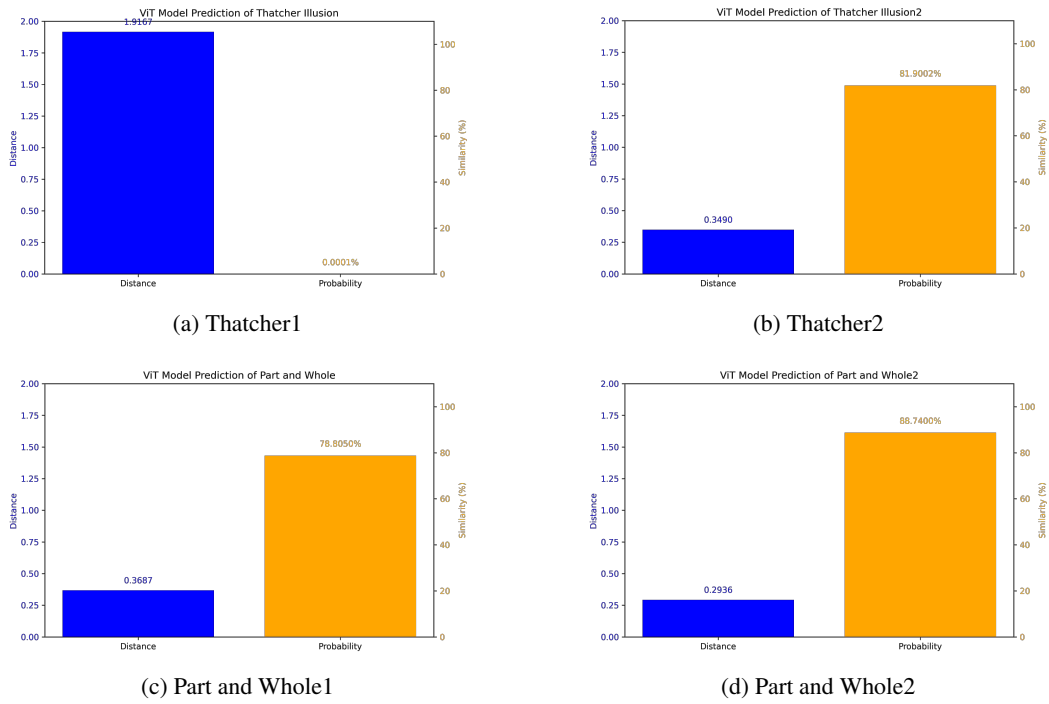


Figure 12: Prediction Result of ViT

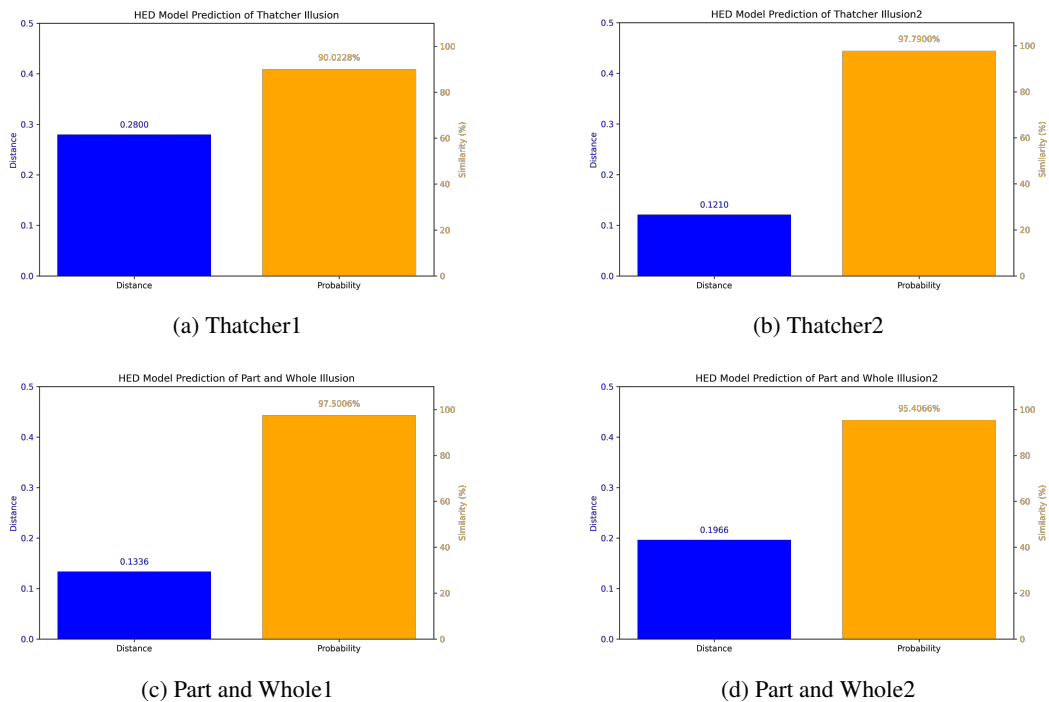


Figure 13: Prediction HED of ResNet50

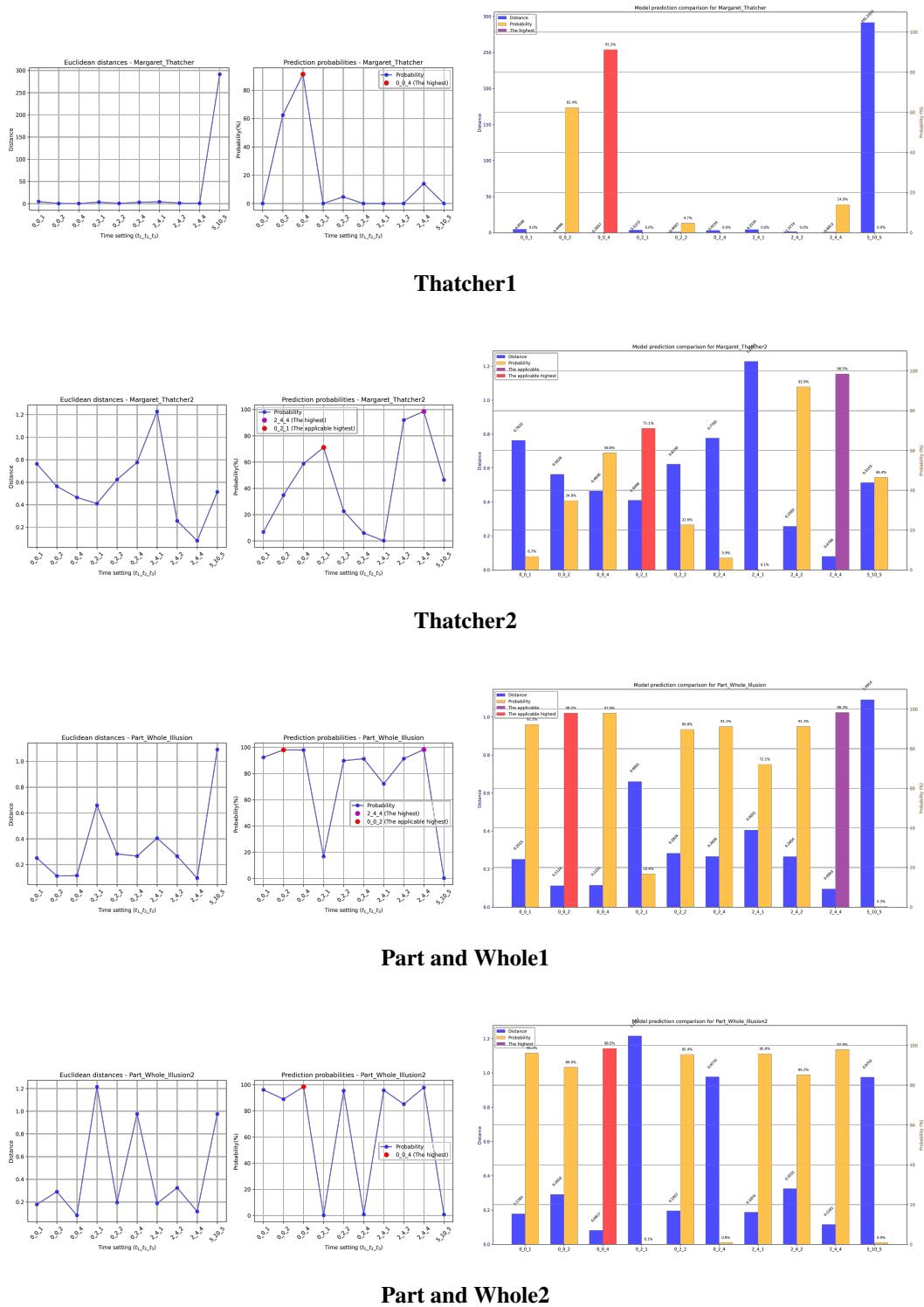
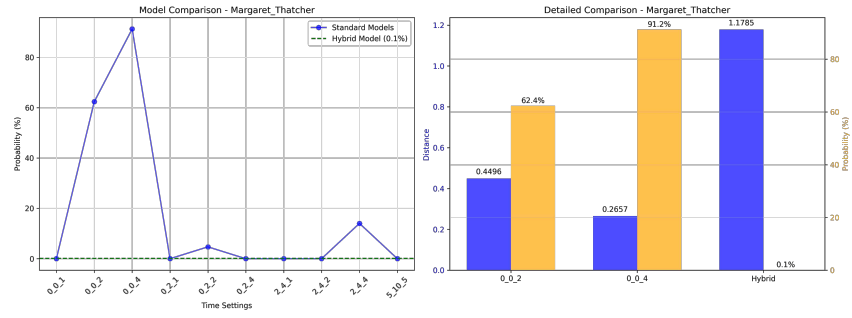
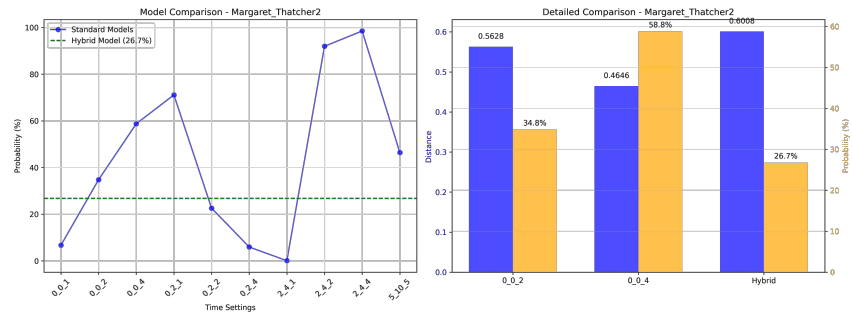


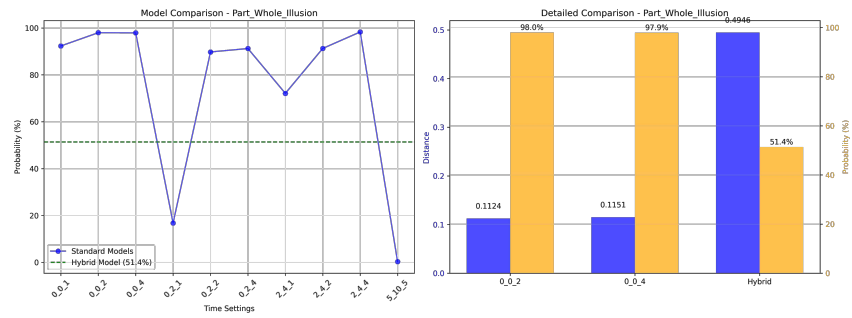
Figure 14: Prediction Result of CORnet-S



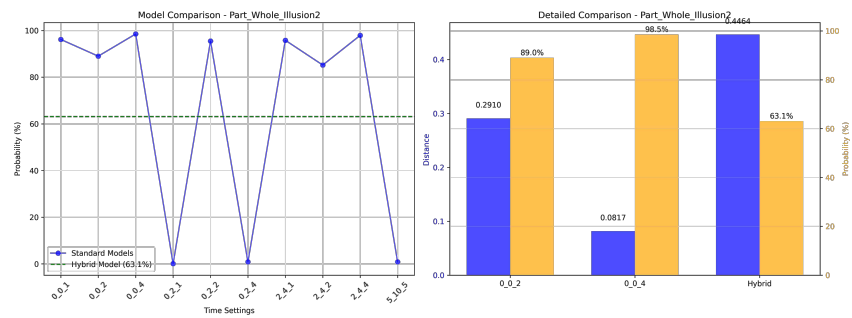
(a) Thatcher1



(b) Thatcher2



(c) Part and Whole1



(d) Part and Whole2

Figure 15: Prediction Result of Hybrid CORnet

B TRAINING CURVES

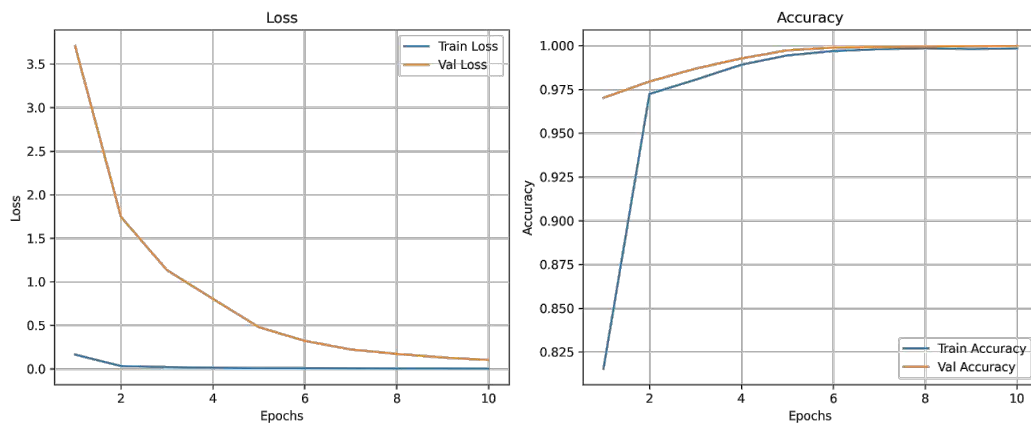


Figure 16: ResNet50 Training Curve

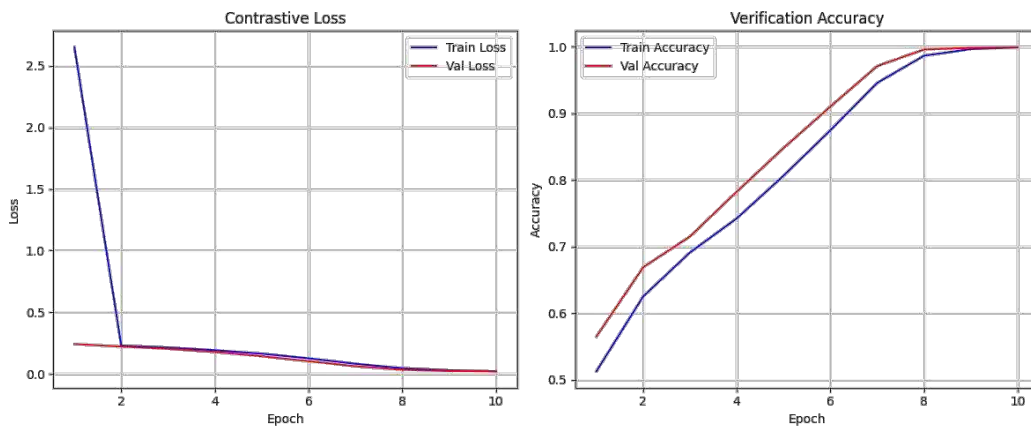


Figure 17: ViT Training Curve

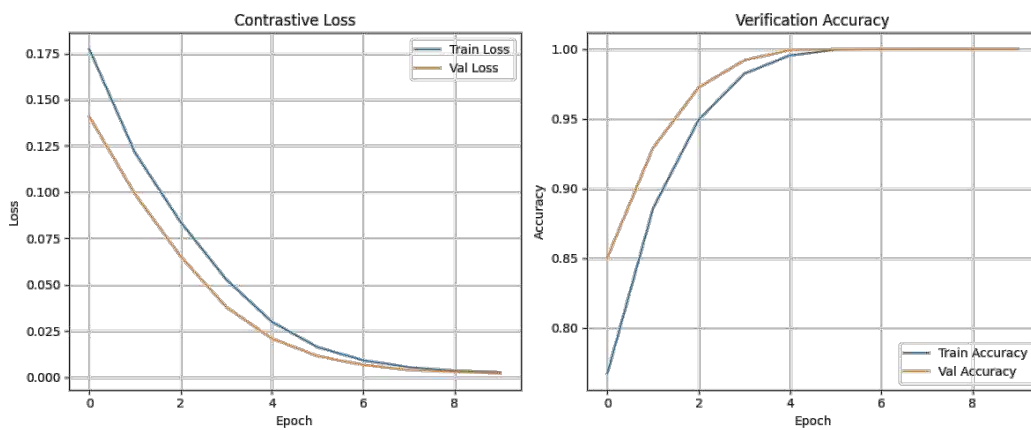


Figure 18: HED Training Curve

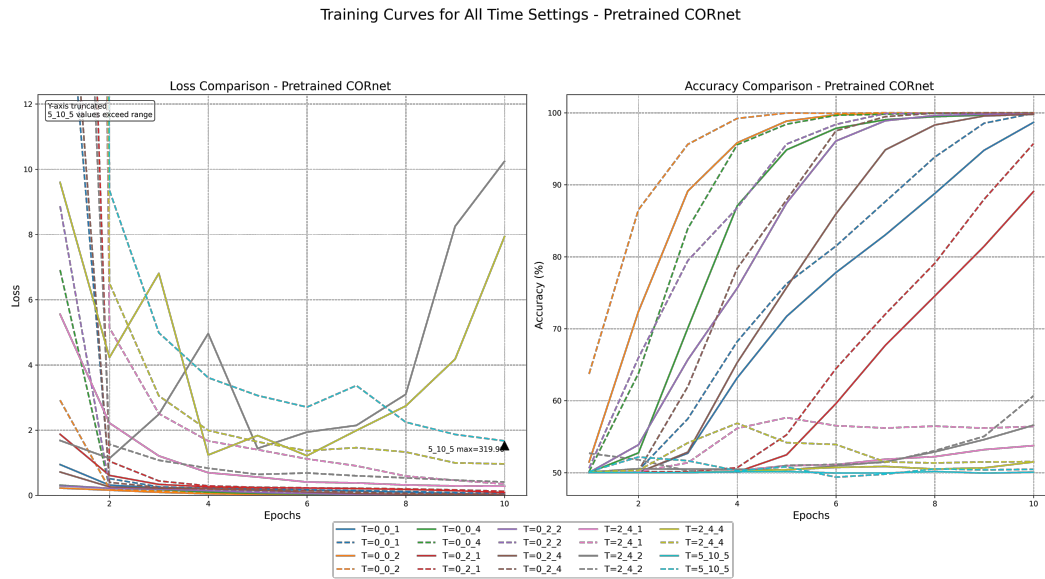


Figure 19: CORnet-S models Training Curve

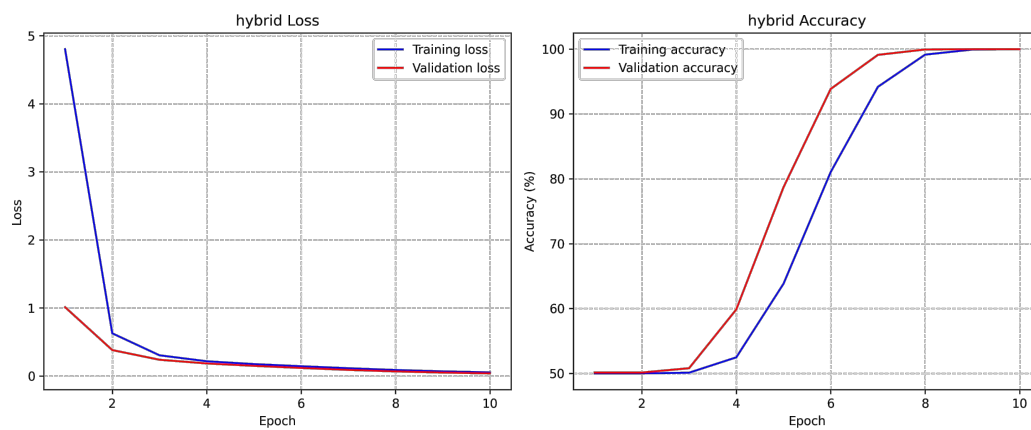


Figure 20: Hybrid CORnet Training Curve

C SALIENCY MAPS

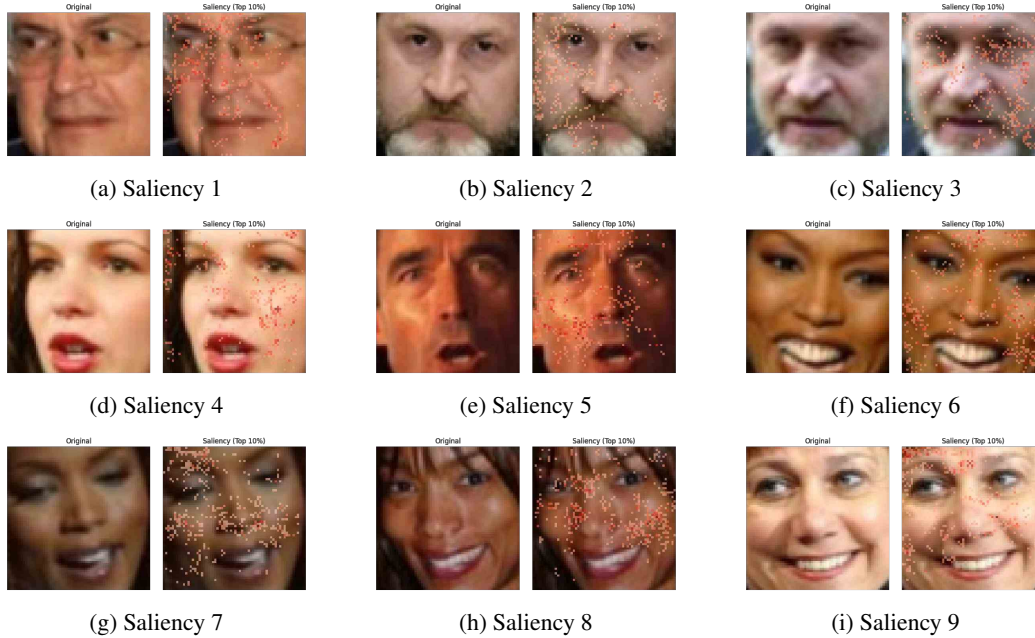


Figure 21: ResNet50 Saliency Map

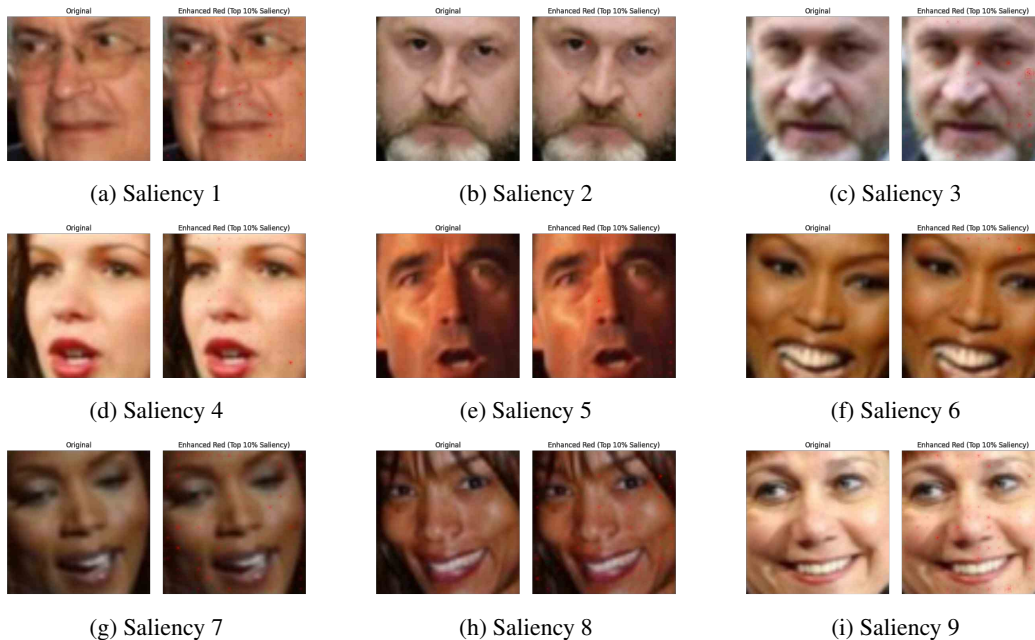


Figure 22: ViT Saliency Map



Figure 23: HED Saliency Map

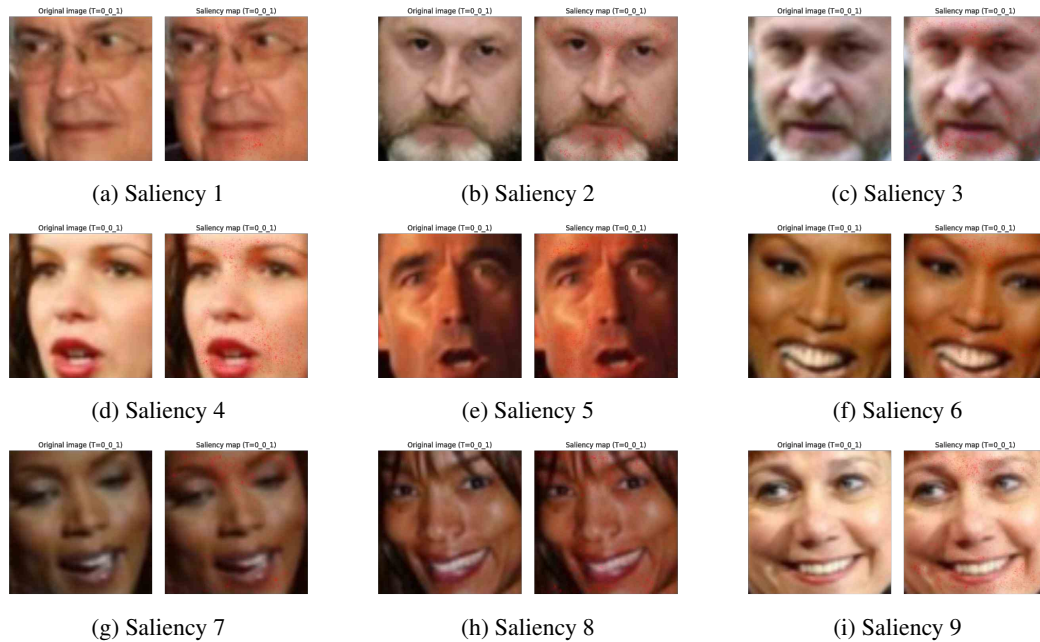


Figure 24: Saliency Map of CORnet-S model with 0.0-1 time settings



Figure 25: Saliency Map of CORnet-S model with 0.0.2 time settings

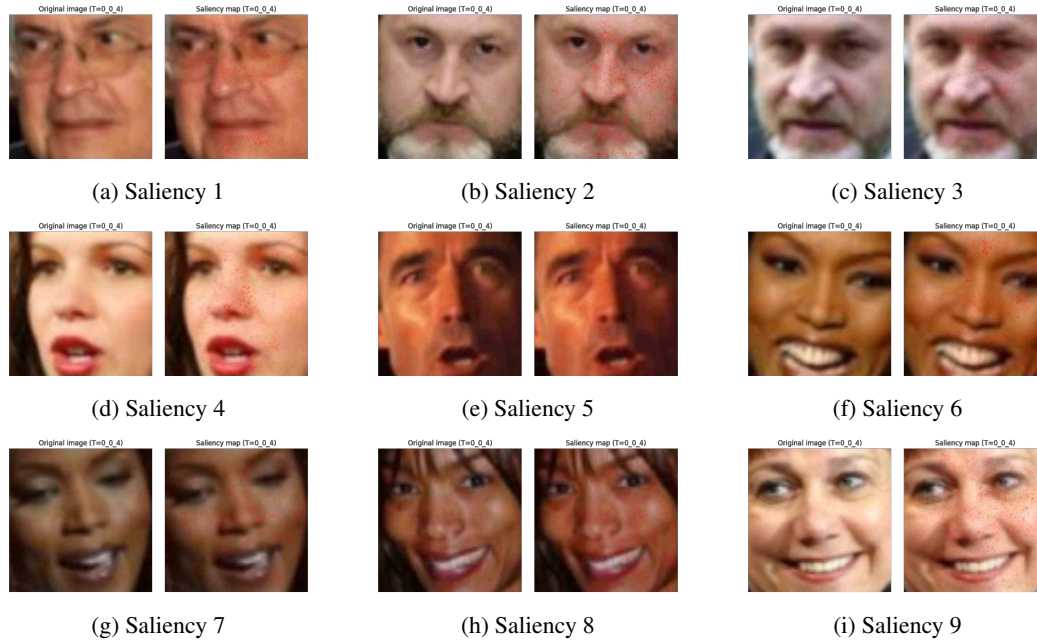


Figure 26: Saliency Map of CORnet-S model with 0.0.4 time settings

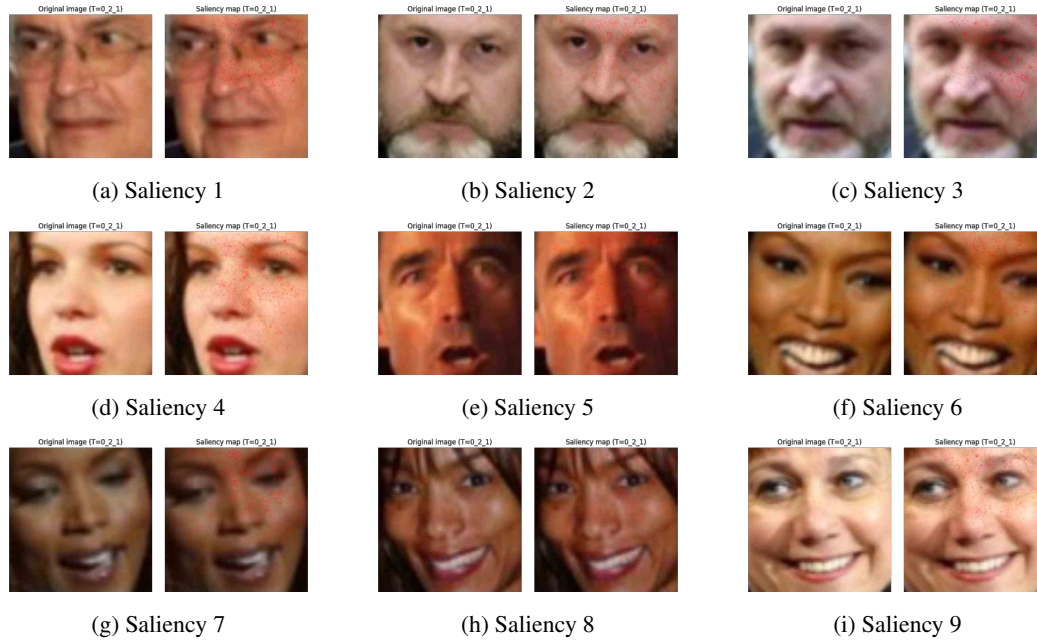


Figure 27: Saliency Map of CORnet-S model with 0.2.1 time settings

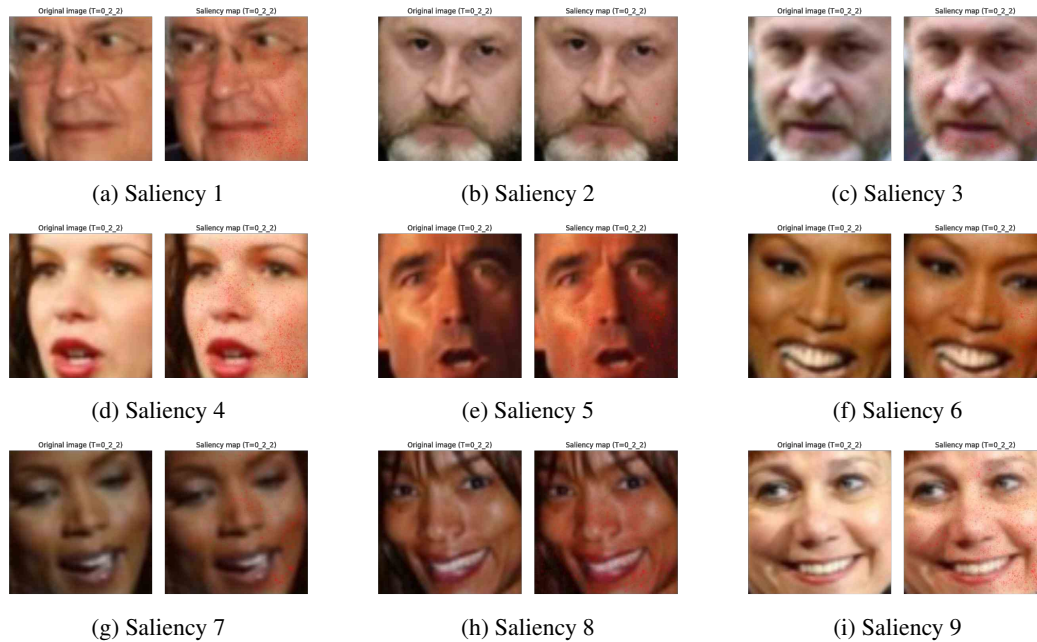


Figure 28: Saliency Map of CORnet-S model with 0.2.2 time settings

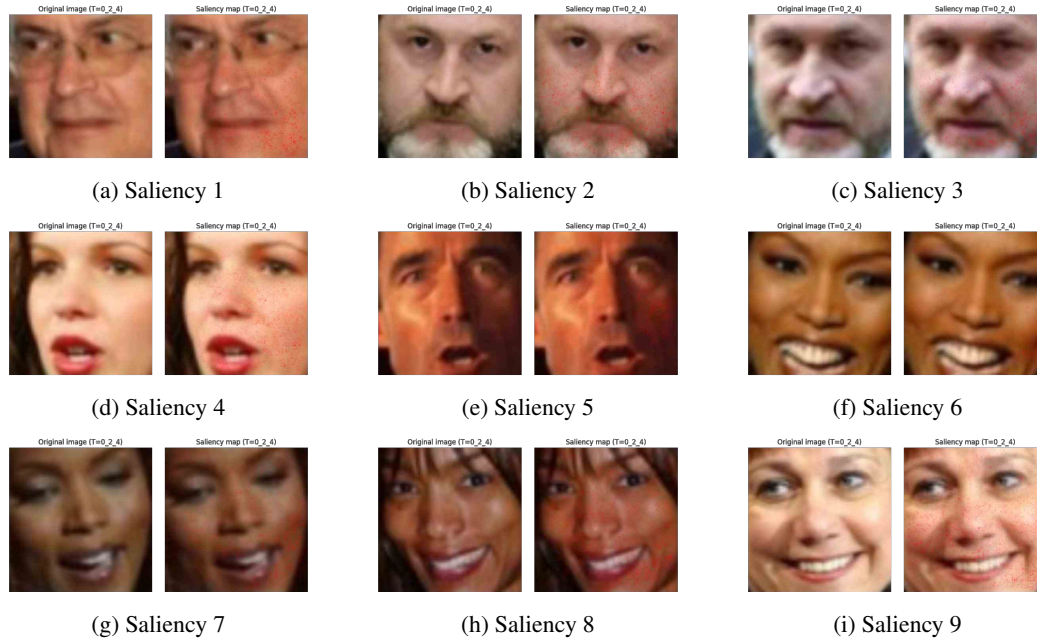


Figure 29: Saliency Map of CORnet-S model with 0.2_4 time settings

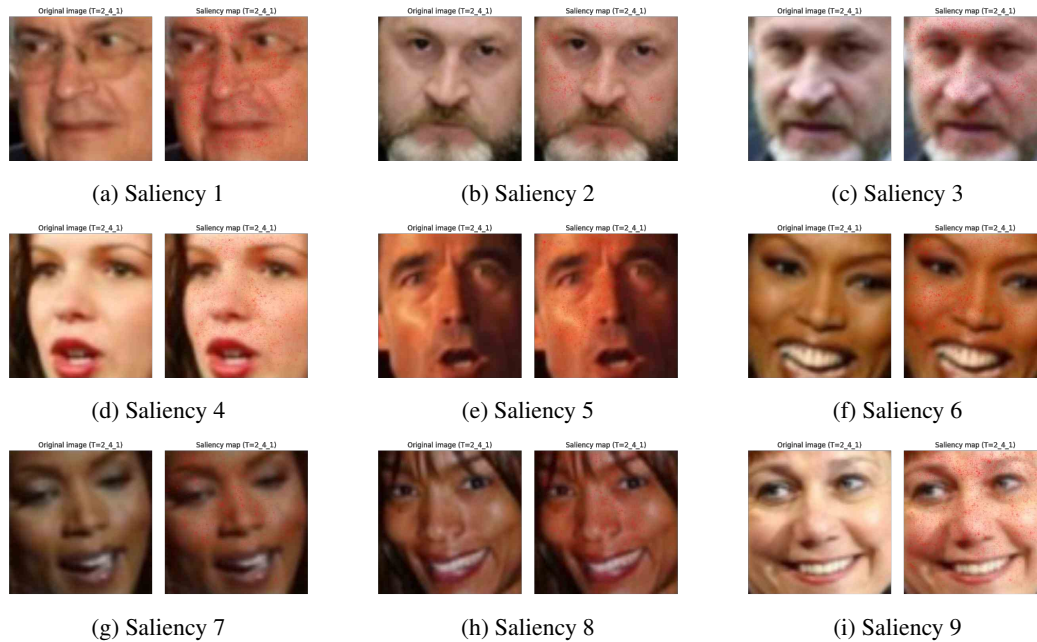


Figure 30: Saliency Map of CORnet-S model with 2.4_1 time settings

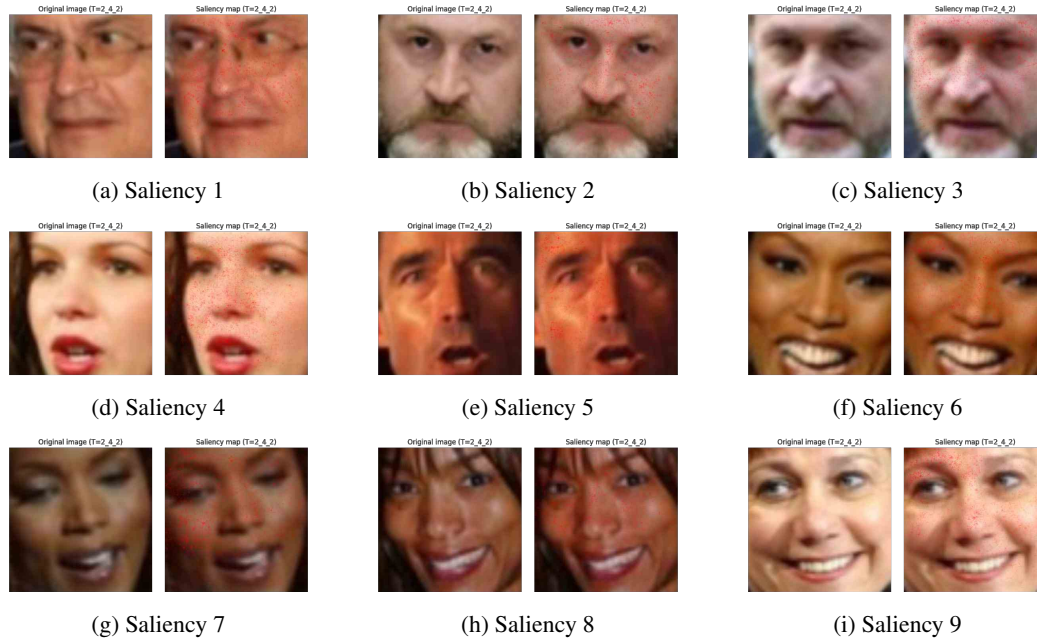


Figure 31: Saliency Map of CORnet-S model with 2.4.2 time settings

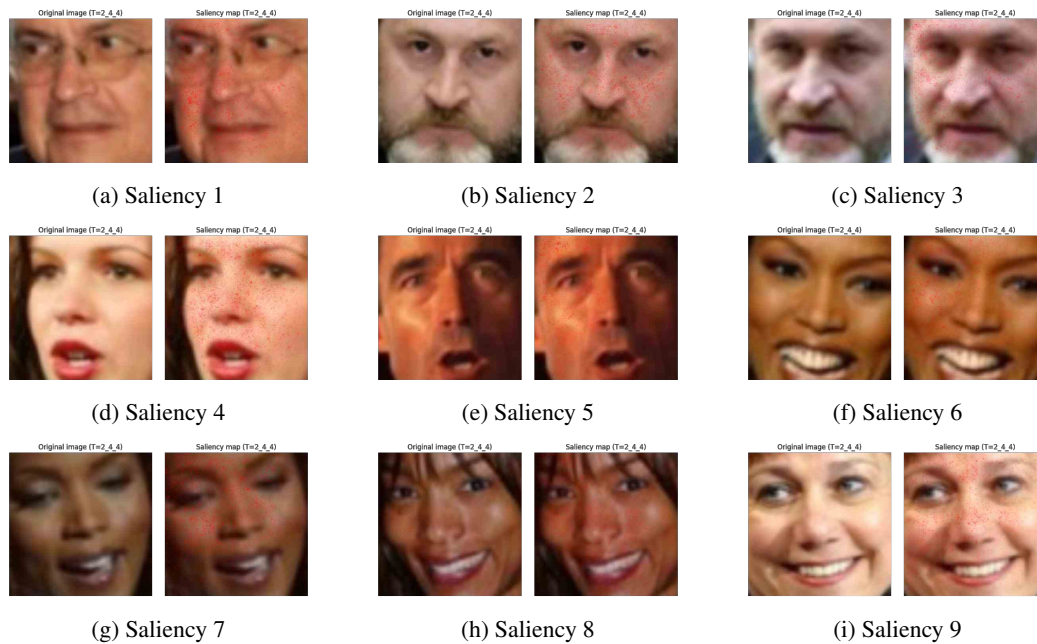


Figure 32: Saliency Map of CORnet-S model with 2.4.4 time settings

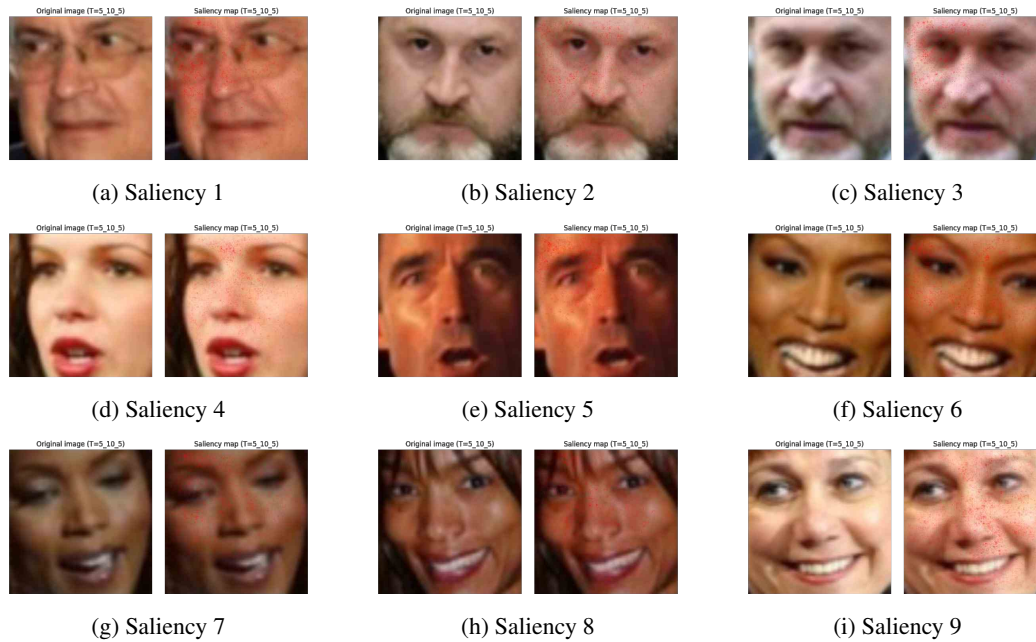


Figure 33: Saliency Map of CORnet-S model with 5_10_5 time settings

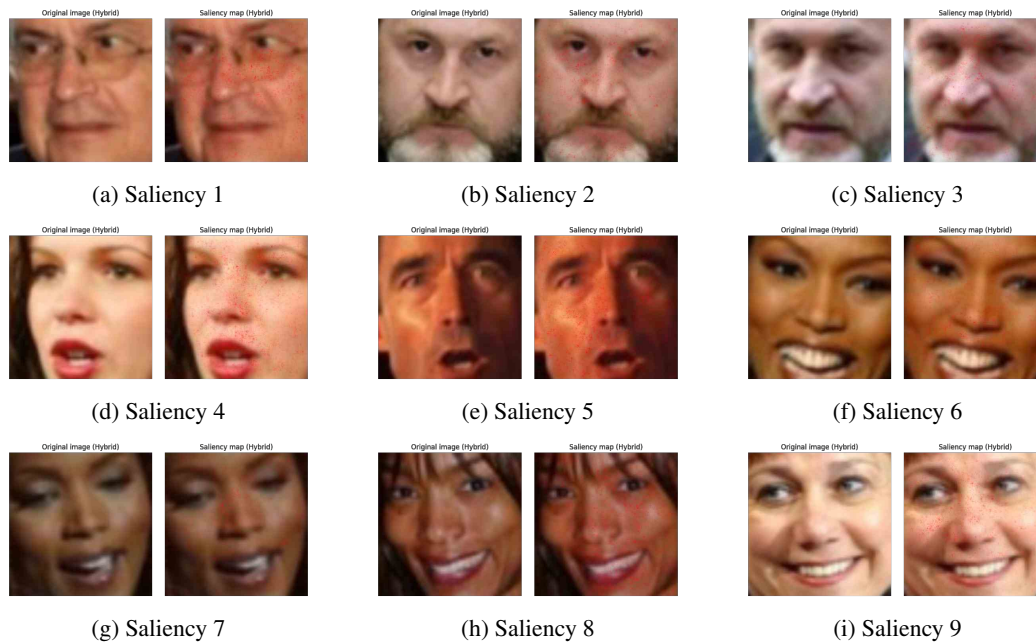


Figure 34: Saliency Map of Hybrid CORnet model

D GEN-AI DECLARATION

I have used ChatGPT and cursor to help me generate some of the code, which is not the key part of the code and which often need to repeat. I use Gen-AI to help me check the grammar mistakes and promote academic writing level in this report in some paragraphs. For every words generated by AI, they are written by myself at first and ChatGPT points out that this word is not so proper in this place, so I follow its suggestions. I finished the design of the project without the help of Gen-AI, and the thought of the whole project is all done by my own.